

Optimizing for Idle Silicon: Correctness-Gated, LLM-Driven Kernel Optimization on the Raspberry Pi 5’s Integrated GPU

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Abstract

Single-board computers ship with integrated GPUs that most edge AI workloads leave idle: inference runs on the CPU and the GPU does nothing. We present a case study in putting that silicon to work. Seppa is a kernel-optimization harness in which a large language model proposes GPU kernel and host-contract changes, and a finite-state machine, not the model, owns compilation, correctness verification, benchmarking, and the keep-or-revert verdict. The machine refuses out-of-order transitions, so a variant that fails verification can never reach a benchmark. Applied to the flood-simulation stencil of bonbibib, an offline flood-aware routing system for the Raspberry Pi 5, the agent-and-gate process produced a fused, strip-mined kernel that is 1.59x faster at 256x256 (about 1,345 to about 2,140 simulation steps per second) with physics verified by three independent gates: numerical agreement with a double-precision CPU reference, mass conservation against injected rainfall, and correct pooling in a terrain basin. The optimization survived an initial false plateau: a first search over shader bodies alone found nothing, because every winning change lived in the host contract (buffer packing, kernel fusion, dispatch geometry). Widening the machine’s action space to include that contract recovered the full speedup, and the machine then re-derived the result end to end over the Model Context Protocol, including refusing to benchmark a deliberately mass-violating kernel. Finally, we measure the point of the exercise: the optimized flood kernel running concurrently with CPU-side language-model inference and street routing. Adding the GPU flood loop to a board saturated with 4-thread LLM decode costs the CPU about 15% of its decode rate (9.5 to 8.1 tokens/s) and buys a continuous 1,100 steps/s simulation; the GPU, sharing one LPDDR memory system with the CPU, retains 52% of its solo throughput. The kernel optimization matters more under contention, not less: its advantage over the original kernel grows from 1.58x alone to 1.84x concurrent, at lower CPU cost. All numbers in this paper carry their exact reproduction commands.

1. Introduction

A Raspberry Pi 5 running a quantized language model uses its four Cortex-A76 cores and leaves its VideoCore VII (V3D) GPU idle. That GPU is modest: 256 maximum invocations per workgroup, 16 KB of shared memory, a fixed subgroup width of 16, no usable fp16 arithmetic, and no cooperative-matrix hardware. It is also, for a workload that does not need the CPU, free.

bonbibib, the motivating application, is an offline flood-aware accessible-routing system for exactly this board [7]. Its division of labor is deliberate: the GPU simulates surface flooding over real terrain with a weighted cellular-automata model (WCA2D [1]), while the CPU concurrently runs a street router (a RoutingKit Customizable Contraction Hierarchy [2]) and a small language model (Granite 4.0 1B, CPU-only llama.cpp [8]) that narrates the computed route. Deterministic code owns the safety-critical logic; the model only phrases it. The GPU flood simulation is what the

idle silicon buys, and its throughput determines how much simulation the system can afford per routing update.

This paper is about how that GPU code got fast, and about the harness that made the process trustworthy. Optimizing kernels for V3D is unforgiving: the driver’s shader compiler falls down a register-allocation fallback ladder on code that any desktop GPU would accept, workgroups over 256 invocations corrupt silently rather than failing loudly, and intuitions imported from CUDA-class hardware (unroll more, prefetch more, tile more) are frequently wrong here, which we know because we measured them being wrong. An optimizer working on this target needs two things: a proposal engine that can absorb unfamiliar constraints quickly, and a verifier that cannot be talked out of the way.

Seppa supplies both, with an explicit separation of powers. A large language model owns only the hypothesize and implement steps of an optimization loop. A Burr [5] finite-state machine owns everything downstream: compile, verify, benchmark, evaluate, `log_variant`. The machine is served over the Model Context Protocol [6], and its single step tool refuses out-of-order calls, so the agent cannot skip verification or read a benchmark for a kernel that failed correctness. Correctness is not a convention the model is asked to follow; it is a transition the machine will not take.

Contributions:

1. A measured account of optimizing a real WCA2D flood stencil for V3D under three physics gates, reaching 1.59x at 256x256, including the falsified hypotheses (Section 4).
2. An action-space lesson for LLM-driven kernel optimization: the first search plateaued at zero improvement not because the kernel was at its limit but because every winning change (buffer packing, fusion, dispatch geometry) lived in the host contract, outside the shader-only action space the machine exposed. Making the host contract part of the implement step’s inputs recovered the full speedup (Section 4.3).
3. A machine-checked reproduction: the FSM, driven over MCP, independently re-derives the result from its own baseline and issues its own keep verdict, and demonstrably refuses to benchmark a kernel that fails the physics gate (Section 5).
4. An interference measurement of the target deployment: the optimized GPU flood loop and CPU LLM decode running concurrently, quantifying what “leveraging idle silicon” costs each side (Section 6).

2. Related work

LLM-driven kernel generation is an active area. KernelBench [3] evaluates models writing CUDA kernels for 250 PyTorch workloads and finds frontier models match the baseline in under 20% of tasks out of the box, improving with execution feedback. FunSearch [4] and AlphaEvolve [9] pair LLM proposal with automated evaluators inside an evolutionary loop, and credit the evaluator, not the model, with the reliability of the results. Seppa shares that architectural conviction and differs in three ways: the target is a hostile, sparsely documented embedded GPU rather than a well-modeled datacenter part; the evaluator is a stateful FSM whose transition structure (not just

its scoring) enforces the discipline, served to the agent as a remote tool it cannot bypass; and the action space includes the host contract, which Section 4.3 shows is where the wins were.

The application side draws on established components: WCA2D [1] for reduced-complexity flood modeling (originally motivated by GPU parallelism), Customizable Contraction Hierarchies [2] for reroutable street routing with millisecond re-customization, and llama.cpp [8] for CPU inference. The V3D driver stack is Mesa’s v3dv [10].

3. Seppa: the machine owns the gate

Seppa is an AutoKernel-style optimization loop restructured as a Burr finite-state machine and served over MCP by a wrapper (Theodosia) running on the Pi itself. The graph is:

```
characterize -> baseline -> hypothesize -> implement -> compile
-> verify -> benchmark -> evaluate -> log_variant -> (hypothesize | stop)
```

with two guard edges that define the system’s character: `compile -> log_variant` on compile failure, and `verify -> log_variant` on gate failure. The benchmark state is reachable only through a green verify. The agent’s entire interface is one MCP tool, `step(action, inputs)`; the server constrains action to the graph’s legal next moves, so “skip verify” is not an expressible request.

The agent owns two states. `hypothesize` returns the current best, the experiment history, and the hardware constraints (256 invocations, 16 KB shared, subgroup 16). `implement` accepts the agent’s proposal. Everything else is deterministic machinery with no model in the loop: compilation via `glslangValidator`, verification and benchmarking via the target harness, and an `evaluate` step that keeps a variant only if it beats the best by more than 1%, persisting the winner to disk. Three consecutive non-improvements, or the experiment budget, ends the run. The machine’s ledger (`variant_log`) records every experiment, kept or reverted, with its gate results.

For the flood target, `verify` runs the physics gate of the `vkflood2` harness: 400 steps at 256x256 must satisfy (a) NMSE below $1e-3$ against a double-precision CPU reference of the same WCA2D update, (b) total water equal to injected rainfall (mass conservation), and (c) maximum depth pooling inside the terrain basin. The benchmark metric is simulation steps per second on the same run. In practice verified NMSE is far below the threshold ($1.3e-9$ at 4,000 steps).

4. Case study: the flood stencil

4.1 The workload

bonbibí’s simulation is a two-pass stencil per time step: a flux pass computes limited inter-cell flows from water-surface-height differences, and a height pass applies net flow plus rainfall to each cell’s depth. On the original harness this ran 400 steps at 256x256 in 0.383 s (about 1,045 steps/s, 1.51 GFLOP/s), with both shaders compiling cleanly. Notably, the shaders are too small to stress the v3dv register allocator, so an optimization playbook proven on llama.cpp kernels for this same GPU (removing source-level unrolling to rescue the compiler’s register allocation, worth +28% on matrix-vector decode there) had nothing to act on here. New target, new bound.

4.2 Falsification sweep

Rather than guess the bottleneck, we bought each hypothesis a variant and let the gate-and-benchmark loop price it. All variants passed all three physics gates; speeds are for 400 steps at 256x256 on the parameterized `vkflood2` harness. Its host loop is itself faster than the original harness for identical kernels (about 1,345 steps/s, 1.94 GFLOP/s, against the 1,045 above), so every comparison in this paper is within `vkflood2`.

Variant	Hypothesis tested	Result
Packed vec2 state (water, surface) halving flux-pass loads	Memory-op count is the bound	1.93 GFLOP/s, no change: falsified
Fused single-dispatch step (flux buffer eliminated)	Traffic and barriers are the bound	2.04 GFLOP/s, +5%: mostly falsified
2 cells per invocation (strip-mined dispatch)	Fixed per-invocation cost is the bound	2.40 GFLOP/s, +23%: supported
4 cells per invocation	More strip is better	2.35 GFLOP/s: falsified, register pressure
Fused + 2 cells per invocation (<code>fused2s.comp</code>)	The wins stack	3.08 GFLOP/s, +59%

The stencil is bound by fixed per-invocation cost: each invocation does so little arithmetic that launch overhead dominates, and amortizing it over two cells is worth more than eliminating half the memory system’s work. Two further V3D specifics shaped the winner. Vertical strips keep a subgroup’s 16 lanes on adjacent x addresses, and lane contiguity matters more on this GPU than per-thread load count. And the kernel had to be written for the register allocator: holding all four flux vectors of the cell pair live fails `v3dv` register allocation and falls down the compiler’s fallback ladder, so the shipped kernel consumes each flux as it is produced and keeps only four scalars.

The shipped kernel is 1.59x at 256x256 (0.187 s vs 0.297 s), 1.41x at 512x512, 1.18x at 1024x1024, and holds all gates over a 4,000-step run (NMSE 1.3e-9, mass conserved, pools in basin). Reproduction:

```
g++ -O3 -o vkflood2 vkflood2.cpp -lvulkan
glslangValidator -V fused2s.comp -o fused2s.spv
STRIP=2 FUSED=1 FLUX_SPV=fused2s.spv ./vkflood2 256 400
```

Integrating it into `bonbib` is three host-side changes: the packed vec2 state buffer, one pipeline and one dispatch per step, and a halved dispatch height (`pi/flood/README.md`).

4.3 The false plateau was an action-space limit

An earlier Seppa session pointed the FSM at this same stencil and measured flat: no variant beat the baseline, and the session concluded there was no headroom. That conclusion was wrong, and the reason matters for anyone building an LLM-driven optimizer. The machine’s implement step accepted only shader source text under a fixed host program. Every change that won in Section 4.2 lives outside that space: vec2 packing is a buffer-layout change, fusion deletes a pipeline and

a barrier from the host loop, and strip-mining changes the dispatch geometry. The search was sound and its result was correct for the space it was allowed to search.

The fix is `v3d_flood2_opt.py`: `implement` now takes `{shader, height_shader, strip}`, where an empty height shader means a fused single-dispatch step and `strip` sets cells per invocation and the dispatch shape. Fusion and dispatch geometry became legal FSM moves rather than out-of-band build decisions, and the machine found and kept the fused strip-2 kernel from its own baseline (Section 5). The general rule we take from this: when a correctness-gated search plateaus, ask whether the winning move is expressible in the action space before concluding the target is exhausted. A plateau is evidence about the search space, not only about the hardware.

5. Machine-checked reproduction over MCP

The optimization claim is only as good as its reproduction path, so the reproduction is executed by the machine, not narrated by the author. `drive_flood2_mcp.py` connects to the Theodosia server's MCP endpoint from a separate machine and drives the full cycle through the step tool, in two acts.

Act one re-derives the result. In the run of 2026-07-11 (transcript in `docs/paper/artifacts/`), the FSM measured its own baseline at 1,346.8 steps/s with all physics gates green, received the fused strip-2 kernel as experiment 1, compiled it, passed all three gates, benchmarked 2,116.4 steps/s, and issued its own verdict: `keep`. That is 1.57x, machine-derived end to end and consistent with the hand-measured sweep. The machine's ledger line reads:

```
{"exp": 1, "fused": true, "strip": 2, "compile_ok": true, "verify_ok": true,
  "steps_per_sec": 2116.4, "best_sps": 2116.4, "verdict": "keep"}
```

Act two demonstrates the gate. The driver submits the same kernel with one change, rainfall injection doubled in one of the two cell updates. It compiles. The physics gate fails it (`verify_ok: false`), and the driver then requests benchmark anyway. The server's verbatim response:

```
{"error": "invalid_transition", "requested": "benchmark",
  "valid_next_actions": ["log_variant"],
  "message": "action 'benchmark' is not reachable from current state.
    Valid actions now: ['log_variant']."}

```

The broken variant enters the ledger as `verdict: revert` with `steps_per_sec: null`. The refused benchmark is the system's thesis in one line: the model, or a buggy or adversarial client, cannot obtain a performance number for physics it broke.

Two independent repetitions agree: an in-process run on the Pi (`replay_flood2.py`, 2026-07-08) measured baseline 1,351 and kept 2,116 steps/s, and a first over-the-wire run earlier on 2026-07-11 measured baseline 1,346.8 and kept 2,105.3 steps/s.

6. The point: concurrent CPU and GPU work

The optimization exists to make a CPU-plus-GPU split worth having, so the last measurement is interference. `bonbibí's` deployment shape is: the GPU loops flood simulation while the CPU runs street routing and Granite 4.0 1B decode (`llama.cpp`, `-ngl 0`, 4 threads; the GGUF reports 1.63 B parameters under `llama.cpp's` granite-3B architecture label, which is what `llama-bench`

prints). The board shares one LPDDR memory system between the halves, so “the GPU is free” is a hypothesis, not a fact, until measured.

`pi/flood/concurrency_bench.sh` measures six conditions on the Pi: each flood kernel alone (4,000-step runs), CPU decode alone (llama-bench tg64, 5 repetitions, 4 threads), the router alone, and each flood kernel looping continuously while the same decode benchmark runs, plus the router under flood load. Every phase starts from below 55 C, and a 1 Hz sampler records temperature and the throttle register throughout; `pi/flood/analyze_conc_bench.py` reduces the raw logs and reports per-phase thermal validity. Concurrent flood runs count only if they start and finish inside the decode window. Results from the run of 2026-07-11 (raw logs in `docs/paper/artifacts/conc_bench/`):

Condition	GPU flood (steps/s)	CPU decode (t/s)
Optimized kernel alone	2,126.9 $\pm$ 0.7 (n=3)	
Original kernel alone	1,348.2 $\pm$ 1.6 (n=3)	
Decode alone		9.5 $\pm$ 0.2
Concurrent, optimized kernel	1,103.1 $\pm$ 145.2 (n=8)	8.1 $\pm$ 0.4
Concurrent, original kernel	599.5 $\pm$ 81.2 (n=6)	7.7 $\pm$ 0.6

The router is not in the table because it does not move: `route.py` completes in 0.02 s alone and 0.02 s under full GPU flood load.

Three results. First, the co-processing thesis holds with a measured price. Adding the optimized flood loop to a board already saturated with CPU decode costs the CPU about 15% of its decode rate and buys a continuous 1,100 steps/s flood simulation that a CPU-only deployment simply does not have. Second, interference is strongly asymmetric: the CPU keeps 85% of its solo throughput, but the GPU keeps only 52%. The two halves share one LPDDR memory system, and the GPU is the victim of that sharing. “Idle silicon” is real but it is not free silicon; it is roughly half a GPU when the CPU is busy. Third, the kernel optimization matters more under contention, not less: the optimized kernel’s advantage over the original grows from 1.58x alone to 1.84x concurrent, and it costs the CPU less (decode retains 8.1 t/s against 7.7 t/s under the original kernel). Both effects are consistent with the fused kernel’s elimination of the intermediate flux-buffer traffic: fewer bytes through the shared memory system per simulation step means less to lose when bandwidth is contended and less pressure exerted on the CPU’s traffic. The optimized kernel is strictly better on both axes of the deployment.

One honesty note on thermals. The 1 Hz traces show that sustained 4-thread decode engages the firmware’s soft temperature limit on this passively cooled board even with no GPU work at all (14 of 34 samples throttled during the decode-alone phase, peak 74.7 C; 29 of 43 and 31 of 45 samples during the concurrent phases, peak 76.3 C). Cool-throughout sustained measurements are unattainable on this cooling, so the numbers above are steady-state, thermally governed figures rather than cold-silicon peaks. For the deployment question this paper asks, that is the right measurement: it is what bonbibí actually gets. The flood-alone measurements, which complete

before heat accumulates, are thermally clean, and the earlier isolated measurements (Section 4) agree with them. Reproduction:

```
OUT=/tmp/conc_bench bash concurrency_bench.sh
python3 analyze_conc_bench.py /tmp/conc_bench 4000
```

7. What transfers and what does not

Three V3D findings from this and companion targets in the same harness, offered as portable heuristics for this class of GPU:

1. Optimize for the driver’s register allocator first. v3dv compiles through a fallback ladder (disabling scheduling, then loop unrolling, then thread count) and the performance cliff is in the ladder, not in your arithmetic. Source-level de-unrolling rescued llama.cpp’s matrix-vector kernel (+28% end-to-end decode); accumulator reduction took a GEMM kernel from 7.02 to 13.42 GFLOP/s in the same FSM under an NMSE gate.
2. Fixed per-invocation cost dominates small kernels. Strip-mining beat every memory-system optimization on the flood stencil, and 2 cells per invocation was the sweet spot before register pressure took the gain back.
3. Lane contiguity beats per-thread operation count. A transposed-B GEMM variant with fewer per-thread loads was verified correct and 36% slower.

The converse also holds: the de-unroll playbook did nothing for the flood stencil because its shaders never stressed the allocator. The harness’s value is exactly that it prices each intuition on each target instead of letting the previous target’s lesson ossify into doctrine.

8. Limitations

The performance envelope is specific: one board (Pi 5, V3D 7.1.10.2, Mesa v3dv 25.0.7), one grid family, and speedups that shrink as the grid grows (1.18x at 1024x1024). The optimized kernel still compiles through part of the v3dv fallback ladder, so headroom likely remains. The concurrency measurement uses llama-bench decode as the CPU load and the raster router as the latency probe; bonbibis’s street-graph CCH re-customization (about 14 ms per flood update on this board) was not separately measured under load. The LLM agent’s proposals were not blind: the fused strip-2 kernel submitted in Section 5’s reproduction was discovered in Section 4’s hand-driven sweep, so the MCP run demonstrates machine verification and verdict, not de novo discovery. And one honest scope note: the same harness’s llama.cpp work on this GPU established that per-op correctness does not compose into end-to-end correctness at full offload due to an upstream driver-independent llama.cpp Vulkan defect we isolated on llvmpipe, so GPU LLM inference on this board currently holds only for a small-offload envelope; that is why bonbibis keeps inference on the CPU and gives the GPU to physics.

9. Reproducibility and availability

Everything is in two repositories: seppa (harness, FSM definitions, MCP server, driver scripts, kernels, and the running notes file docs/fable_notes/NOTES.md with one entry per confirmed win and dead end) and bonbibis (the application). The flood kit is pi/flood/: kernels, parameterized harness vkflood2.cpp, the falsification-sweep variants, and concurrency_bench.sh.

The FSM target is `v3d_flood2_opt.py`, served by `theodosia_server.py --http --flood2; drive_flood2_mcp.py` reproduces Section 5 against that endpoint, and `replay_flood2.py` does the same in-process on the Pi. Raw logs for Section 6 are produced by `concurrency_bench.sh` into a directory of plain-text files, from which every number in the table derives.

10. AI involvement

The optimization agent in this work is a large language model (Anthropic Claude, Table 5 for the experiments reported here) operating through Seppa’s MCP interface; that is the system under study, not an authorship aside. The same class of model assisted in drafting this paper’s text. All measurements were produced by the deterministic harness described above, and every kept kernel passed machine-owned verification gates that the model could not bypass.

References

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